

MEASUREMENT MCQ (9th standard)

1. Measurement quantifies object characteristics by comparing with
- a) similar objects
 - b) standard quantity
 - c) mathematical formula
 - d) visual estimation

Correct answer: b

2. Physical quantities independent of others called
- a) derived quantities
 - b) fundamental quantities
 - c) vector quantities
 - d) scalar quantities

Correct answer: b

3. SI unit system developed by
- a) United Nations
 - b) General Conference Weights Measures 1960
 - c) British Standards Institution
 - d) American Society Testing Materials

Correct answer: b

4. Metre defined as distance light travels vacuum
- a) $1/299792458$ second
 - b) $1/3000000000$ second
 - c) exactly 1 second
 - d) 1000 seconds

Correct answer: a

5. Astronomical unit measures distance

- a) Earth-Moon
- b) Earth-Sun average
- c) Sun nearest star
- d) within galaxy

Correct answer: b

6. Light year equals distance light travels

- a) one day
- b) one month
- c) one year vacuum
- d) one parsec

Correct answer: c

7. 1 parsec equals

- a) 1 light year
- b) 3.26 light years
- c) 9.46 trillion km
- d) 1 AU

Correct answer: b

8. Angstrom unit measures

- a) astronomical distances
- b) atomic distances
- c) mass
- d) time

Correct answer: b

9. Atomic mass unit defined as 1/12 mass

- a) hydrogen-1 atom

b) carbon-12 atom

c) oxygen-16 atom

d) helium-4 atom

Correct answer: b

10. Kelvin defined as fraction thermodynamic temperature

a) boiling point water

b) $1/273.16$ triple point water

c) absolute zero

d) ice point

Correct answer: b

11. SI prefixes deca means

a) 10^{-1}

b) 10^1

c) 10^2

d) 10^3

Correct answer: b

12. Unit symbol named after scientist written

a) small letters

b) capital letters

c) italics

d) underlined

Correct answer: b

13. Newton second written as

a) Ns

b) N·s

c) N/s

d) NS

Correct answer: b

14. Metre per second correctly written

a) m/s

b) ms^{-1}

c) m s^{-1}

d) both a b

Correct answer: d

15. Vernier calliper least count typically

a) 1 mm

b) 0.1 mm

c) 0.01 mm

d) 0.001 mm

Correct answer: b

16. Vernier calliper measures with accuracy

a) 1 mm

b) 0.1 mm

c) 0.01 mm

d) 1 cm

Correct answer: b

17. Main scale vernier calliper graded

a) mm inches

b) cm mm

c) m cm

d) only mm

Correct answer: a

18. Positive zero error occurs when zero reading

- a) above actual zero
- b) below actual zero
- c) coincides zero
- d) negative value

Correct answer: a

19. Actual reading formula zero error

- a) $MS + VS$
- b) $MS + VS - ZE$
- c) $MS - VS + ZE$
- d) $MS \times VS$

Correct answer: b

20. Screw gauge least count

- a) 0.1 mm
- b) 0.01 mm
- c) 1 mm
- d) 0.001 mm

Correct answer: b

21. Screw gauge pitch typically

- a) 0.5 mm
- b) 1 mm
- c) 0.1 mm
- d) 10 mm

Correct answer: b

22. Head scale divisions screw gauge

- a) 50

b) 100

c) 10

d) 200

Correct answer: b

23. Ratchet screw gauge prevents

a) zero error

b) over tightening

c) under tightening

d) slipping

Correct answer: b

24. Thickness formula screw gauge

a) $PS + (HS \times LC)$

b) $PS + HS$

c) $HS \times PS$

d) $PS - HS$

Correct answer: a

25. Common balance compares mass using

a) spring extension

b) equal arm balance

c) digital display

d) magnetic field

Correct answer: b

26. Physical balance sensitivity

a) 5g

b) 1 mg

c) 1 g

d) 10 mg

Correct answer: b

27. Digital balance least count

a) 1 g

b) 10 mg

c) 1 mg

d) 100 mg

Correct answer: c

28. Spring balance works Hooke's law

a) $F = kx$

b) $F = ma$

c) $PV = nRT$

d) $E = mc^2$

Correct answer: a

29. Mass differs from weight because mass

a) changes location

b) measure of inertia

c) gravitational force

d) both b c

Correct answer: b

30. Accuracy defined as closeness to

a) average value

b) true value

c) repeated readings

d) instrument limit

Correct answer: b

31. Precision concerns

- a) true value
- b) reproducibility
- c) absolute error
- d) calibration

Correct answer: b

32. Derived quantity example

- a) length
- b) force
- c) mass
- d) time

Correct answer: b

33. Force unit newton equals

- a) kg m s^{-1}
- b) kg m s^{-2}
- c) $\text{kg m}^2 \text{s}^{-2}$
- d) m s^{-2}

Correct answer: b

34. Volume SI derived unit

- a) m^3
- b) cm^3
- c) litre
- d) ml

Correct answer: a

35. Density equals mass per

- a) volume

- b) area
- c) length
- d) time

Correct answer: a

36.Speed unit

- a) m s
- b) m/s
- c) ms^{-1}
- d) b c same

Correct answer: d

37.Pressure unit pascal

- a) N m^{-1}
- b) N m^{-2}
- c) kg m s^{-2}
- d) both b c

Correct answer: d

38.Unit prefix tera means

- a) 10^{12}
- b) 10^9
- c) 10^6
- d) 10^3

Correct answer: a

39.Pico prefix equals

- a) 10^{-9}
- b) 10^{-12}
- c) 10^{-6}

d) 10^{-3}

Correct answer: b

40. Vernier scale 10 divisions equal

a) 9 main scale divisions

b) 10 main scale divisions

c) 11 main scale divisions

d) 8 main scale divisions

Correct answer: a

41. Zero error positive when zero mark

a) coincides

b) right of index

c) left of index

d) damaged

Correct answer: c

42. Screw gauge positive zero error head scale zero

a) coincides pitch

b) below pitch axis

c) above pitch axis

d) damaged

Correct answer: b

43. Beam balance standard masses up to

a) 1 kg

b) 5 kg

c) 10 kg

d) 100 g

Correct answer: b

44. Physical balance measures accurately

- a) gram
- b) milligram
- c) kilogram
- d) tonne

Correct answer: b

45. Instrument measures wire diameter

- a) vernier calliper
- b) screw gauge
- c) beam balance
- d) spring balance

Correct answer: b

46. Depth measurement uses

- a) outer jaws
- b) inner jaws
- c) depth rod
- d) vernier scale

Correct answer: c

47. Anvil screw gauge opposite

- a) spindle
- b) ratchet
- c) sleeve
- d) pitch scale

Correct answer: a

48. Mass constant while weight varies due

- a) location gravity

b) instrument error

c) temperature

d) pressure

Correct answer: a

49. Scientific notation density mercury

a) 13600 kg m^{-3}

b) $1.36 \times 10^4 \text{ kg m}^{-3}$

c) $13.6 \times 10^3 \text{ kg m}^{-3}$

d) $1.36 \times 10^3 \text{ g cm}^{-3}$

Correct answer: b

50. Unit symbols never

a) plural

b) end full stop

c) capital letters

d) separated space

Correct answer: b

51. Ampere written as

a) amp

b) A

c) ampere

d) Am

Correct answer: b

52. Least count = smallest division main scale /

a) vernier divisions

b) total divisions

c) pitch

d) zero error

Correct answer: a

53. Digital vernier advantage

a) manual calculation

b) direct reading display

c) cheaper

d) heavier

Correct answer: b

54. Spring balance measures

a) mass

b) weight

c) volume

d) density

Correct answer: b

55. Accuracy improves by

a) larger readings

b) smaller readings

c) averaging

d) calibration only

Correct answer: b

56. Precision high accuracy

a) always high

b) always low

c) may low

d) impossible

Correct answer: c

57. Parsec used for

- a) solar system
- b) interstellar distances
- c) atomic scale
- d) time measurement

Correct answer: b

58. 1 AU approximately

- a) 1.5×10^8 km
- b) 9.46×10^{12} km
- c) 3.26 ly
- d) 1 ly

Correct answer: a

59. Temperature scales Celsius Kelvin relation

- a) $K = ^\circ C + 273.15$
- b) $K = ^\circ C - 273.15$
- c) $K = ^\circ C \times 1.8$
- d) $K = ^\circ C / 1.8$

Correct answer: a

60. Luminous intensity SI unit

- a) lux
- b) lumen
- c) candela
- d) watt

Correct answer: c

61. Electric current fundamental quantity unit

- a) volt

b) ampere

c) ohm

d) watt

Correct answer: b

62.Amount substance measured

a) grams

b) moles

c) litres

d) kilograms

Correct answer: b

63.Vernier calliper inner jaws measure

a) external diameter

b) internal diameter

c) depth

d) thickness

Correct answer: b

64.Screw gauge ratchet clicks indicate

a) zero error

b) proper pressure

c) slipping

d) damage

Correct answer: b

65.Common balance limitation

a) measures milligrams

b) compares equal arms

c) digital display

d) spring mechanism

Correct answer: b

66. Physical balance standard masses

a) 1 mg to 200 g

b) 5 g to 5 kg

c) 1 g to 100 g

d) 10 g to 1 kg

Correct answer: a

67. Unit prefix giga equals

a) 10^6

b) 10^9

c) 10^{12}

d) 10^3

Correct answer: b

68. Mega prefix

a) 10^3

b) 10^6

c) 10^9

d) 10^{12}

Correct answer: b

69. Micro equals

a) 10^{-6}

b) 10^{-3}

c) 10^{-9}

d) 10^{-12}

Correct answer: a

70. Nano equals

- a) 10^{-6}
- b) 10^{-9}
- c) 10^{-12}
- d) 10^{-3}

Correct answer: b

71. Vernier coincidence means

- a) main vernier division align
- b) zero error
- c) least count
- d) pitch

Correct answer: a

72. Screw gauge backlash error due

- a) ratchet
- b) loose screw
- c) worn threads
- d) b c

Correct answer: d

73. Instrument sheet thickness

- a) vernier calliper
- b) screw gauge
- c) beam balance
- d) spring balance

Correct answer: b

74. Jar depth measured

- a) outer jaws

b) vernier scale

c) depth rod

d) lock screw

Correct answer: c

75. Mass standard kilogram preserved

a) Paris

b) Washington

c) London

d) Moscow

Correct answer: a

76. Platinum-iridium cylinder kilogram prototype

a) 1799

b) 1889

c) 1960

d) 1983

Correct answer: b

77. Redefinition 2019 kilogram uses

a) Planck constant

b) Avogadro number

c) light speed

d) atomic mass

Correct answer: a

78. Error systematic affects

a) precision

b) accuracy

c) both

d) neither

Correct answer: b

79. Random error affects

a) accuracy

b) precision

c) calibration

d) zero setting

Correct answer: b

80. Significant figures vernier reading 2.34 mm has

a) 2

b) 3

c) 4

d) 5

Correct answer: b